

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
```

```
data=pd.read_csv("scores.csv")
data
```

	Hours_studied	Final_score
0	1	50
1	2	55
2	3	65
3	4	70
4	5	75
5	6	78
6	7	85
7	8	90

```
x=pd.DataFrame(data['Hours_studied'])
y=pd.DataFrame(data['Final_score'])
```

```
plt.xlabel('Hours studied')
plt.ylabel('Final scores')
plt.scatter(data.Hours_studied,data.Final_score,color='red',marker='^')
```

<matplotlib.collections.PathCollection at 0x22e27901f10>



```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2)
```

```
linm=LinearRegression()
```

```
linm.fit(x_train,y_train)
```

```
▼ LinearRegression  
LinearRegression()
```

```
linm.intercept_
```

```
array([46.6])
```

```
linm.coef_
```

```
array([[5.55]])
```

```
linm.predict(x_test)
```

```
array([[79.9],  
       [57.7]])
```

```
y_test
```

	Final_score
5	78
1	55

```
linm.predict([[9]])
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\base.py:464: UserWarni  
warnings.warn(  
array([[96.55]])
```

